CANDIDATE CASE STUDY 04

Environmental data that can survive an audit

Carbon accounting, ISO 14001:2026 readiness and GRI disclosure plan

Evidence boundary. The assessment supplied a five-page brief but no operational CSV. The companion Python notebook uses 2,000 generated records to demonstrate the controls. Every generated figure and factor is illustrative. This report therefore makes no claim about an actual company footprint, certification status or peer ranking.

Executive decision

Approve a 30-day evidence sprint before publishing totals or scheduling a certification audit. First reconcile facility activity and source documents. Then approve site/year emission factors and eligible electricity instruments. Finally test the ISO controls and publish a GRI content index with explicit omissions.

Accounting design

Ledger	Calculation	Control
Scope 1	Reported direct tCO2e; fuel-level reconstruction once quantities and factors arrive	Reject negative or missing values. Require fuel, refrigerant and source evidence.
Scope 2 location	Grid kWh x geographic grid kg CO2e/kWh / 1,000	Report gross figure separately from market method.
Scope 2 market	Eligible kWh x contractual factor + remainder x residual/grid factor	An offset alone does not reduce gross Scope 2. Verify instrument quality and vintage.
Scope 3 partial	Purchased water service (Cat 1) + third-party waste treatment (Cat 5)	Screen all 15 categories. Label the present estimate partial.

Example quality gates: trim and standardize site names; normalize audit-status labels; quarantine recycling shares above 100%, negative Scope 1, impossible renewable claims and duplicate IDs; retain each raw value and correction history. Do not silently cap or reverse invalid values.

ISO 14001:2026 | audit gap and action matrix

Statuses below are **Not assessed** because the brief provides no site evidence. “Gap signal” states what the described manual trackers leave unproven, not a finding of nonconformity. Confirm clause wording against the controlled standard copy before audit.

Area / clause	Evidence to request	Gap signal from brief	Action / owner / timing
Context & scope / 4	EMS boundary, site list, interested-party needs	Global facilities with hybrid trackers	Approve operating boundary and site register / EHS lead / week 1
Leadership & policy / 5	Policy approval, roles, responsibilities	No policy evidence supplied	Confirm policy, site accountability / EHS director / week 1
Aspects & impacts / 6.1.2	Register with significance method and lifecycle view	Resource and waste data lack aspect link	Score water, energy, waste, spills / site EHS / week 2
Compliance obligations / 6.1.3	Legal register, permits, periodic evaluations	No jurisdiction-specific register supplied	Map permits and evaluation dates / legal + EHS / week 2
Objectives / 6.2	Baseline, targets, owners and monitored plan	No approved baseline or targets supplied	Set measurable objectives after data lock / EHS / week 3
Operational control / 8.1	Procedures, contractor controls, records	Manual trackers may lack sign-off	Version controls and owner review / operations / week 3
Emergency response / 8.2	Scenario plan, drills, incident records	No drill evidence supplied	Run spill scenario and record learning / site EHS / week 3
Monitoring & audit / 9.1-9.3	Calibration, internal audit, management review	No documented review cycle supplied	Test samples and close findings / audit lead / week 4

Certification gate: release the audit pack only after evidence owners sign off the matrix, legal evaluations are complete, and internal audit findings have documented corrective actions.

GRI benchmark | disclosure readiness

The benchmark is completeness against the relevant GRI disclosures, followed by comparable year-on-year intensity when scope, boundaries and denominators are stable. GRI does not supply one universal “good” carbon or recycling target.

Topic	Relevant disclosures	Current evidence position	Next data request
Climate	GRI 305-1 to 305-3; transition to GRI 102	Partial model only; Scope 3 incomplete	Gas/source detail, factor provenance, category screen
Energy	GRI 302; transition to GRI 103	Electricity present; other fuels unknown	Fuel and purchased heat, renewable instrument file
Water	GRI 303-3 to 303-5	Consumption field only	Withdrawal source, discharge and water stress
Waste	GRI 306-3 to 306-5	Mass and recycling share only	Composition and treatment/disposal route

Repeatable tracker and review cadence

Monthly site upload: record ID, reporting period, site, source unit, quantity, supporting invoice/meter/manifest, owner and approval date. The central EHS analyst runs validation and factor joins; site owners clear exceptions; finance or sustainability signs off the locked ledger. Quarterly, track data coverage, quarantined records, factor provenance, Scope 1, both Scope 2 methods, screened Scope 3 categories, water intensity, and waste diverted by verified route.

30-day delivery sequence

Week 1: obtain the actual CSV and source documents, agree boundary and site registry. **Week 2:** resolve quarantined records, build factor register and aspect/legal registers. **Week 3:** verify contractual instruments, complete waste and water evidence, test emergency and operational controls. **Week 4:** perform internal sample audit, approve the carbon ledger, issue the GRI index and management action log.

Standards and method references

ISO 14001:2026: https://www.iso.org/standard/14001
GHG Protocol Scope 2 Guidance: https://ghgprotocol.org/sites/default/files/GHG%20Protocol%20Scope%202%20Guidance.pdf
GRI standards library: https://www.globalreporting.org/how-to-use-the-gri-standards/gri-standards-english-language/
GRI 102 and 103 take effect for reporting from 1 January 2027:
https://www.globalreporting.org/standards/standards-development/topic-standard-for-climate-change-and-energy/